EmpathAI: Multi-Signal Mental Health Detection with Retrieval-Grounded Support

Ramya Sri Sushma Vasamsetti[1] **Paavani Ramakrishna**[2] **Harsh Kelawala**[3]
Mansi Suryawanshi[4] Shruti Dhamecha[5]
University of Southern California, USA

[1]rvasamse@usc.edu [2]paavanir@usc.edu [3]kelawala@usc.edu [4]msuryawa@usc.edu [5]sdhamech@usc.edu

Abstract

Online mental health forums contain rich signals about users' psychological states, yet manually triaging thousands of daily posts is infeasible. EmpathAI is an NLP pipeline that detects mental-health signals in Reddit posts and generates grounded, safety-aware responses. The system addresses two tasks: multi-label intent detection across nine categories and single-label concern severity classification (low, medium, high). We compare a MiniLM + Logistic Regression baseline with transformer models fine-tuned using LoRA (Hu et al., 2021), including DistilRoBERTa, RoBERTa-large, and DeBERTa-v3. A retrieval-augmented generation module (Lewis et al., 2020) grounds responses in counselor-style examples, while a multi-tier safety layer appends crisis resources for high-risk posts. RoBERTa-large achieves the best intent classification (Macro-F1: 0.774) and concern classification (Accuracy: 0.814), and the RAG pipeline attains a grounding score of 0.70, demonstrating that combining classification, retrieval, and safety-aware generation yields a reliable and responsible mental-health NLP system.

1 Introduction

Online support communities such as r/depression and r/SuicideWatch see thousands of posts daily from users describing emotional distress, harmful coping patterns, and urgent requests for help. These posts contain signals critical for identifying a user's psychological state and the urgency of their need. Yet manually reviewing this volume of content is infeasible, and delayed responses to high-risk posts carry serious consequences (De Choudhury et al., 2013; Coppersmith et al., 2015).

Keyword-based triage systems fail to capture context: the phrase "I want to die" may signal genuine crisis or everyday frustration, and only deeper understanding can tell the difference. Most existing automated approaches address either crisis risk or topical classification, but not both simultaneously (Matero et al., 2019). Furthermore, even when detection is accurate, generating responses without grounding them in verified professional guidance risks producing clinically inappropriate or harmful advice.

EmpathAI addresses these gaps through a unified modular pipeline. We frame the problem as two complementary tasks: multi-label intent detection, since a single post can simultaneously express distress, identify a cause, and request help; and single-label concern severity prediction to enable urgency-based triage. For response generation, we retrieve counselor-style examples from a curated knowledge base and condition generation on retrieved content, keeping classification and generation strictly separated for safety. A multi-tier crisis detection layer ensures that high-risk posts receive appropriate safety resources.

Our key results show that RoBERTa-large with LoRA and Focal Loss achieves a Macro-F1 of 0.774 for intent classification and an accuracy of 0.814 for concern classification, outperforming all baselines. The RAG pipeline achieves an average retrieval similarity of 0.71 and a grounding score of 0.70, and the safety layer successfully triggers crisis resource injection for high-risk posts.

2 Related Work

Chancellor and De Choudhury (2020) reviewed predictive techniques for mental health on social media, identifying two persistent gaps: most systems address only one dimension of risk, and nearly none include safe response generation after detection. EmpathAI directly addresses both by combining multi-label intent classification with grounded response generation and crisis intervention.

Inamdar et al. (2023) applied ELMo and BERT embeddings to classify Reddit posts as stressful versus non-stressful, achieving an F1 of 0.76. While effective, the binary framing cannot capture that a single post may simultaneously express crisis risk, identify a cause, and request help, a limitation our nine-category multi-label formulation addresses.

Hu et al. (2021) proposed LoRA, which freezes pretrained weights and injects trainable low-rank matrices into transformer layers, matching full fine-tuning performance while reducing trainable parameters by orders of magnitude. We adopt LoRA across all transformer models to enable efficient training on our modest and unbalanced data set. To further handle class imbalance, we incorporate Focal Loss (Lin et al., 2017), which down-weights well-classified examples and focuses optimization on hard minority-class instances, a technique not used in prior mental health classification work.

Lewis et al. (2020) introduced retrieval-augmented generation, showing that grounding language model outputs in retrieved documents reduces hallucination over purely generative approaches. However, their work focused on answering open-domain questions without domain-specific safety constraints. Unlike prior RAG systems, our generation pipeline incorporates multi-tier crisis detection and harmful content filtering, requirements unique to the mental health domain.

3 Problem Description

We define two complementary tasks on a dataset of Reddit posts drawn from mental health support communities. **Intent Classification** is formulated as a multi-label problem, since a single post may simultaneously express multiple psychological signals. Each post is assigned one or more labels from nine categories: Cause of Distress, Critical Risk, Maladaptive Coping, Mental Distress, Miscellaneous, Mood Tracking, Positive Coping, Progress Update, and Seeking Help. **Concern Classification** is formulated as a single-label problem, where each post is assigned exactly one severity level, Low, Medium, or High, to enable urgency-based triage.

The dataset consists of approximately 6,000 Reddit posts with significant class imbalance, with Mental Distress and Medium concern being the dominant classes. A gold set of 500 posts was manually annotated; weak supervision (confidence ≥ 0.85) and a BART-large-MNLI zero-shot labeling pipeline expanded the dataset to $\sim$8,000 posts. All experiments use a fixed stratified split of 3,694 training, 792 validation, and 792 test samples, managed via a central YAML configuration for full reproducibility.

4 Methods

MiniLM + Logistic Regression (Baseline). Posts are encoded into 384-dimensional vectors using `all-MiniLM-L6-v2`. A One-vs-Rest Logistic Regression with balanced class weights handles intent (threshold = 0.5), and a multinomial Logistic Regression predicts concern level. No GPU is required; training takes $\sim$2 seconds.

DistilRoBERTa + LoRA. We fine-tune `distilroberta-base` (82M parameters) with LoRA on all 6 attention layers (rank = 64, $\alpha = 128$, dropout = 0.1), yielding 1.77M trainable parameters (2.1%). Intent uses weighted Binary Cross-Entropy; concern uses class-weighted cross-entropy. Trained for 10 epochs, batch size 8, learning rate 1e-4 with cosine decay. Threshold optimization on validation yields an optimal intent threshold of 0.34.

RoBERTa-Large + LoRA + Focal Loss. We fine-tune `roberta-large` (355M parameters) with LoRA on all 24 layers (rank = 32, $\alpha = 64$), yielding 14.2M trainable parameters (3.8%). Focal Loss (Lin et al., 2017) ($\gamma = 2.0$) is used for intent to emphasize hard minority-class examples. All other hyperparameters match DistilRoBERTa.

DeBERTa-v3 + LoRA. An additional benchmark using DeBERTa-v3's disentangled attention with LoRA fine-tuning, introduced to assess whether enhanced attention improves multi-label classification over RoBERTa variants.

RAG Pipeline. The generation pipeline (Lewis et al., 2020) operates in three phases. In

Model	Macro F1	Micro F1
MiniLM + LR	0.6956	0.7330
DistilRoBERTa + LoRA	0.6459	0.6417
DeBERTa-v3 + LoRA	0.7260	0.7705
RoBERTa-Large + LoRA	**0.7740**	**0.8200**

Table 1: Intent classification performance. See Figure 1 in Appendix.

Model	Accuracy	Macro F1
MiniLM + LR	0.7247	0.7084
DistilRoBERTa + LoRA	0.7652	0.7805
DeBERTa-v3 + LoRA	0.6755	0.6564
RoBERTa-Large + LoRA	**0.8144**	**0.7857**

Table 2: Concern-level classification performance. See Figures 2 and 3 in Appendix.

Metric	Value
Retrieved Snippets	3
Average Similarity	∼0.71
Grounding Score	0.70
Crisis Detected	Yes

Table 3: RAG evaluation metrics. See Figure 4 in Appendix.

the *retrieval phase*, the input post is encoded with `all-MiniLM-L6-v2` and queried against a FAISS index of 50,172 counselor response snippets tagged with intent and concern metadata; the top-3 most similar snippets are retrieved and filtered by predicted labels. In the *generation phase*, retrieved snippets and the input post are passed to Flan-T5-large (max 400 tokens); responses are accepted only if grounding ratio $\geq$ 0.70. In the *safety phase*, a multi-tier crisis detection module appends crisis resources (988 helpline; text HOME to 741741; findahelpline.com) when high-risk content is detected, and logs all high-risk interactions to a daily JSONL file.

5 Experimental Results

5.1 Intent Classification Results

The intent classification task is formulated as a multi-label problem over nine categories, where each post may simultaneously express multiple psychological signals such as mental distress, help-seeking, or coping behavior. This makes the task inherently challenging due to label overlap and semantic similarity between categories.

RoBERTa-large achieves the best performance across both Macro-F1 and Micro-F1 metrics (Table 1, Figure 1), demonstrating its effectiveness at capturing both overall classification accuracy and balanced performance across all labels including minority classes. It improves Micro-F1 by approximately **5% over DeBERTa-v3** and nearly **9% over the MiniLM baseline**, highlighting the benefit of larger transformer architectures for multi-label understanding. DeBERTa-v3 emerges as the second-best model, while DistilRoBERTa shows the weakest performance due to its reduced parameter size limiting its ability to model subtle overlapping signals. Overall, the results demonstrate a clear trend: **increasing model capacity significantly improves multi-label classification performance**.

5.2 Concern Classification Results

Concern classification is modeled as a single-label classification task where each post is assigned a severity level: Low, Medium, or High.

RoBERTa-large again achieves the highest performance (Table 2, Figure 2), outperforming DistilRoBERTa by approximately **5%** and MiniLM by nearly **9%** in accuracy. Notably, DeBERTa-v3 underperforms all models on this task despite its strong intent results, indicating that model rankings are task-dependent. The confusion matrix (Figure 3) reveals that most errors occur at the Medium-High boundary, where severity depends on subtle emotional intensity rather than explicit lexical cues. This highlights a key challenge: **severity classification requires deeper contextual and emotional understanding beyond surface-level text patterns**.

5.3 Retrieval-Augmented Generation (RAG) Results

The RAG component extends the system beyond classification by generating grounded, context-aware responses based on retrieved knowledge.

The retrieval module achieves an average similarity of ∼0.71 between posts and retrieved snippets, indicating strong semantic alignment. The grounding score of 0.70 confirms that generated responses are closely anchored to retrieved knowledge, reducing hallucination risk in this safety-critical domain. Generated responses were qualitatively observed to be empathetic, relevant, and actionable. The safety layer successfully triggered crisis resource injection for high-risk posts, with all interactions logged for audit.

6 Conclusion and Future Work

EmpathAI demonstrates that a unified pipeline combining multi-label intent classification, severity prediction, retrieval-augmented generation, and safety-aware intervention produces a reliable and responsible mental-health NLP system. RoBERTa-large with LoRA and Focal Loss consistently achieves the strongest classification results across both tasks, and the RAG pipeline generates grounded responses that reduce hallucination risk. A key remaining challenge is the Medium-High concern boundary, where subtle emotional cues make classification difficult even for large models, and where errors carry the greatest real-world consequences.

For future work, we plan to expand manual annotations to reduce label noise and revisit overlapping intent categories. To improve sensitivity to high-risk posts, we will apply class-aware sampling and targeted augmentation focused on the Medium–High boundary. We will also explore hierarchical or label-aware architectures to model dependencies between co-occurring intent labels, and expand the counselor knowledge base to improve RAG response quality and coverage across diverse communities and writing styles.

Individual Contributions

Ramya Sri Sushma Vasamsetti: Collected and cleaned the Reddit dataset, coordinated manual annotation via Label Studio, and resolved inter-annotator disagreements. Introduced the DeBERTa-v3 + LoRA model and contributed to FAISS indexing of counselor responses.

Paavani Ramakrishna: Defined the annotation schema for intent and concern labels, wrote the tag normalization module, and set up stratified data splits. Implemented the BART-large-MNLI zero-shot labeling pipeline and built the knowledge base construction pipeline including chunking and embedding of counselor responses.

Harsh Kelawala: Built all classification models (MiniLM + LR, DistilRoBERTa + LoRA, RoBERTa-Large + LoRA) with Focal Loss and threshold optimization. Retrained models on the BART-augmented dataset and implemented the Flan-T5 generation module within the RAG pipeline.

Mansi Suryawanshi: Ran training experiments and conducted error analysis with per-label F1 breakdowns and confusion matrices. Built the RAG evaluation framework covering relevance, grounding, safety, and crisis coverage scoring, and compiled final comparative results.

Shruti Dhamecha: Designed the YAML-based configuration system and GitHub repository structure for full reproducibility. Implemented the multi-tier crisis detection system, unsafe content filter, and safety resource routing. Built the interaction logging infrastructure and engineered the batch generation pipeline with timeout protection and resumable processing.

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A Figures

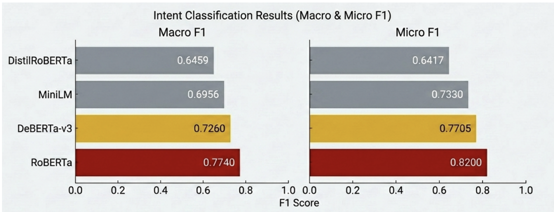

Figure 1: Macro-F1 and Micro-F1 comparison across models for intent classification.

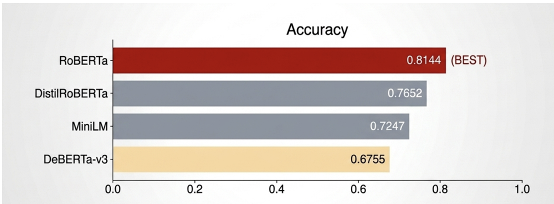

Figure 2: Accuracy comparison across models for concern classification.

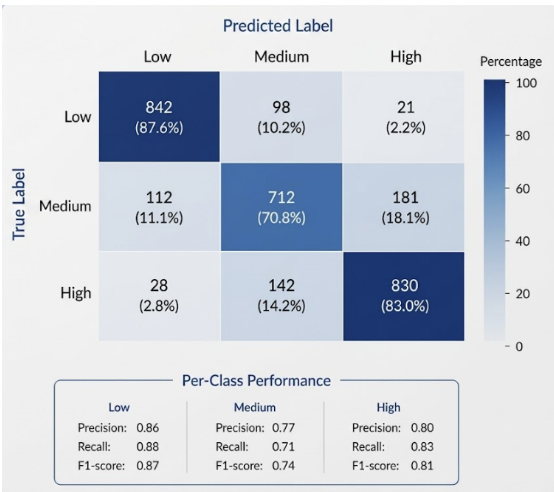

Figure 3: Confusion matrix for concern classification.

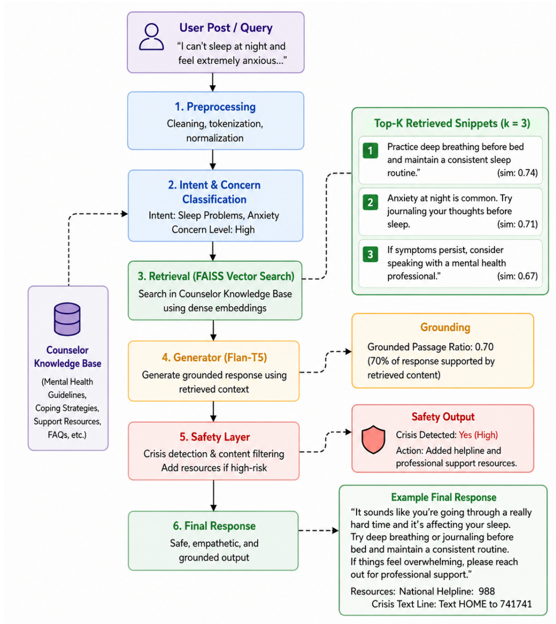

Figure 4: Retrieval-Augmented Generation pipeline showing retrieval and response generation flow.